WEBB INDUSTRIES' COMPRESSED-AIR IC ENGINE ACHIEVES 110.5 W MECHANICAL AT 0.7 V WITH 355°–45° INJECTION AND 4 FLYWHEELS; CROSS-TDC INJECTION

AUTHORS
KATHERINE CIAVOLA
JOAO FONSECA
GARY PARKER

RECIPIENT
WEBB INDUSTRIES,
ATTN: JULIE CHANG

DATE
NOVEMBER 6, 2025

CLASS & SECTION
ME 4056, A01

SUPERVISORS
DAVID MACNAIR
JILL FENNEL

- CHASE SUN
- AAVASH BUDHATHOKI

BURDELL, INC.
ANALYTICAL DIVISION

Introduction

Webb Industries commissioned Burdell, Inc. to assess and configure a compressed-air IC engine coupled to a 48 V regenerative motor for the purpose of charging a battery. To inform choices for maximum power generation, a model of the system was created by developing a calibration procedure, quantifying frictional effects and experimental flywheel inertia, validating the work done by the air per stroke, and establishing an injection strategy that maximizes shaft power. Using a linear York regression on force-transducer data, the calibration equation is described by $T = G\,V_{ft} + T_0$, with $G = 0.717 \pm 0.002\ Nm/V$ and $T_0 \approx -0.003 \pm 0.05\ Nm$ .Steady-state characterization supports a Newtonian friction behavior $T_f(\omega) = 0.0040\omega + 0.4286\ Nm$ for a given input speed in rad/s. Acceleration tests give an experimental flywheel inertia (I_{fly}) of $2.8 \times 10^{-3} \pm 3.046 \times 10^{-5}\ kg \cdot m^2$ (close to the $2.6 \times 10^{-3} \pm 3.12 \times 10^{-5}$ design value). With the head installed (no air injection), segmenting the data into a representative cycle confirm physically consistent signs. However, applying an adiabatic assumption (≈-22.05 J for the power/compression stroke) reveals a small modeling gap attributable to leakage. Enabling angle-based air injection resulted in a peak power output of ~110.5 W with an injection window of 355°–45°. Operation at $\omega \approx 90 - 110\frac{rad}{s}$ (860 – 1050 rpm) with 3–4 flywheels is recommended for stable power generation. To align prediction and measurement, introducing a better sealing method is also recommended; this change directly tightens the energy balance and makes the timing map actionable for control.

Methodology

Burdell Inc. employed a work-energy model to (i) establish a repeatable torque calibration procedure, (ii) identify mechanical losses due to friction and experimental flywheel inertia with the head removed, (iii) identify stroke-resolved air-work with the head installed (no injection), and (iv) map crank-angle injection for maximum output power. All analyses are referenced to the 720° cycle and reported with coverage factor K = 2 unless otherwise specified.

Calibration of the force transducer converts the NI ELVIS III voltage, V_{ft} to torque through $T = G\,V_{ft} + T_0$. During calibration, the motor angle is set to top dead center (TDC) to limit offset variance due to motor angle. Known masses are then applied at a measured lever arm, L, to generate reference torques (T_{cal} = mgL) over multiple loading and unloading levels to adjust for hysteresis in the transducer. Uncertainty in T_{cal} is obtained by partial derivatives with respect to m and L, combining instrument accuracy and digital resolution (rectangular PDFs for calipers and mass) and treating ELVIS quantization as triangular with accuracy added in quadrature. A York regression (errors in both axes, r = 0) provides G and T_0 with standard errors; parameter covariances are retained for later propagation. Calibration is performed at the start of each session and repeated when drift or re-mounting occurs; all subsequent torque data are computed with the most recent calibration. Calibration figures include the best-fit line with K = 2 bands.

Encoder data (4000 counts/rev) are binned at 10° to form θ_k buckets; angular velocity $\omega(\theta)$ is computed from count differentials and averaged within each bucket to suppress quantization noise. For each trial, cycles are unwrapped and re-indexed so that top-dead-center aligns consistently; the minimum of $\omega(\theta)$ per revolution is used to refine the TDC marker. Steady-state windows are identified in the tail of each record when the running mean of ω remains within ±5% over at least five consecutive buckets; pre-steady segments provide acceleration data for transient characterization. A representative cycle is constructed by averaging $T(\theta)$ and $\omega(\theta)$ over at least ten complete steady cycles. Nominal stroke boundaries at 0°, 180°, 360°, 540°, and 720°

are applied, with a ±10° boundary adjustment permitted when the derivative $d\omega/d\theta$ indicates a clearer physical transition; the same boundaries are then used for torque and energy integration.

With the head off, mechanical loss is modeled from steady points as a function of speed. The baseline form is a combined Coulomb-plus-Newtonian model, $T_f(\omega) = b + D\omega$, expressed in rad/s. Using the chosen friction form, flywheel inertia for each plate stack is estimated from the pre-steady acceleration segment via the work-energy theorem,

$$\frac{1}{2}\, I_{fly}(\omega_b^2 - \omega_a^2) = \sum\, (T_{motor}(\theta) - T_f\,(\omega(\theta)))\,\Delta\theta,\ \ \Delta\theta =\ 10^o \cdot \frac{\pi}{180}\ \ [1].$$

Multiple runs per configuration are processed to assess repeatability; the session-consistent I_{fly} is then fixed for each inertia setting during head-on analyses.

With the head installed and no injection, stroke-resolved air-work is identified to validate the model and establish a theoretical reference. For each stroke (s) of the representative cycle,

$$W_{air,s} = \Delta E_{fly,s} + W_{motor,s} - W_{fric,s}\ [2a]$$

$$\Delta E_{fly,s} = \frac{1}{2} I_{fly}(\omega^2(\theta_{out}) - \omega^2(\theta_{in}))\ \ [2b]$$

where $W_{\text{motor},s} = \sum T(\theta)\Delta\theta$ and $W_{\text{fric},s} = \sum T_f(\omega(\theta))\,\Delta\theta$ over stroke s. A thermodynamic bound for compression/expansion is computed using the measured bore, stroke, and head clearance to form $V(\theta)$, assuming dry-air, isentropic behavior; this model is used only as a reference to quantify discrepancy. Cycle-closure checks enforce $\sum \Delta E_{\text{fly},s} \approx 0$ at steady speed; any residual trend triggers a review of segmentation and friction selection.

Injection mapping is performed with the pneumatic valve enabled so that the engine drives the motor. The injection profile is parameterized by center c (ATDC) and width w. For a grid of (c, w) at several speeds and flywheel stacks, the net work to the motor per cycle,

$$W_{\to motor} = \sum_{\theta=0}^{720^o} T(\theta)\,\Delta\theta\ \ \ [3],$$

is computed from calibrated torque, and mechanical power is obtained by $P = W_{\to\text{motor}} \times (\text{rpm}/120)$. The resulting $P(c, w, \omega)$ surfaces are used to identify timing that maximizes power, while maintaining manageable torque ripple; recommended operating windows are extracted from these maps in the Results.

Uncertainty is propagated at three levels. First, calibration parameters (G, T_0) and DAQ voltage uncertainty enter each torque sample through linearization with the York covariance. Second, per-bin work terms combine torque and angle uncertainties: u_θ follows from encoder count resolution mapped to radians and bucketization; u_T combines calibration and sample statistics. Third, stroke and cycle aggregates combine per-bin contributions in quadrature and include between-cycle variability from the representative cycle averaging. Friction and inertia parameter covariances are propagated into W_{fric} and ΔE_{fly}, respectively. All models are fit and reported in rad/s, quality gates reject runs with non-monotone encoder counts or inadequate steady data, and figures in Results pair model overlays with $K = 2$ bands to make assumptions and prediction limits explicit.

Results

The investigation produced a calibration procedure, a model of frictional effects, verified with experimental inertia, stroke-resolved air-work to quantify leakage, and an optimal compressed-air injection window that delivers generator-useful power.

System Identification and Torque Scale (York Regression)

Because the motor shaft is physically unbalanced, the calibration procedure was performed at top dead center (TDC) before each trial was started. In addition to an angular dependance, hysteresis in the torque transducer creates a discrepancy between loading and unloading, as shown in Figure 1.

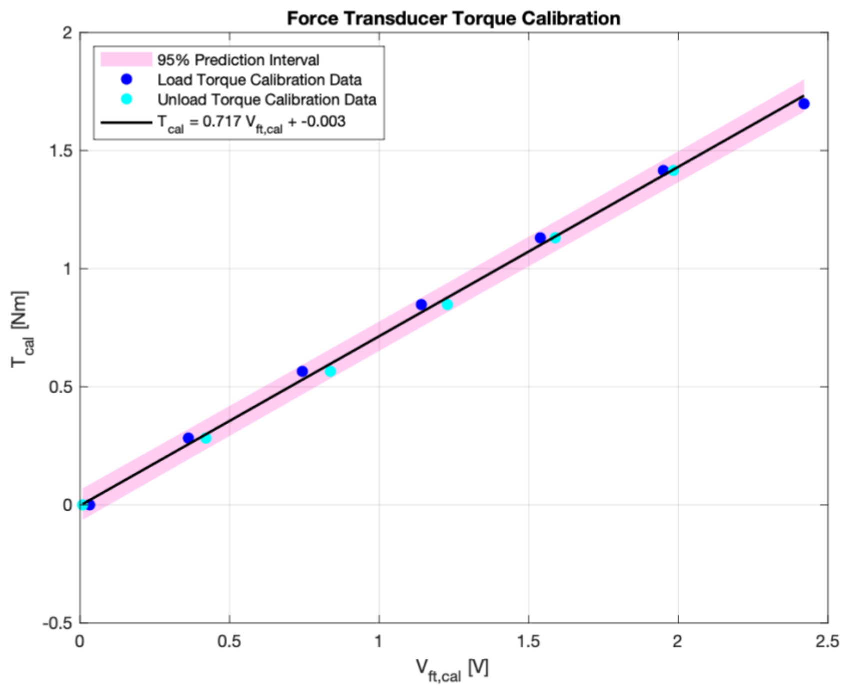

Figure 1: *Torque calibration T vs. transducer voltage V_{ft} with York fit and K=2 prediction band*

This discrepancy introduces variability in the measured torque between trials. To determine a repeatable calibration procedure and quantify downstream effects on work calculations, a new calibration was recorded before each run. However, as shown in Figure 2, re-calibrating the system before each trial did not improve the discrepancy between experimental and theoretical values for the work done by air; there is no “best” calibration method to reduce uncertainty.

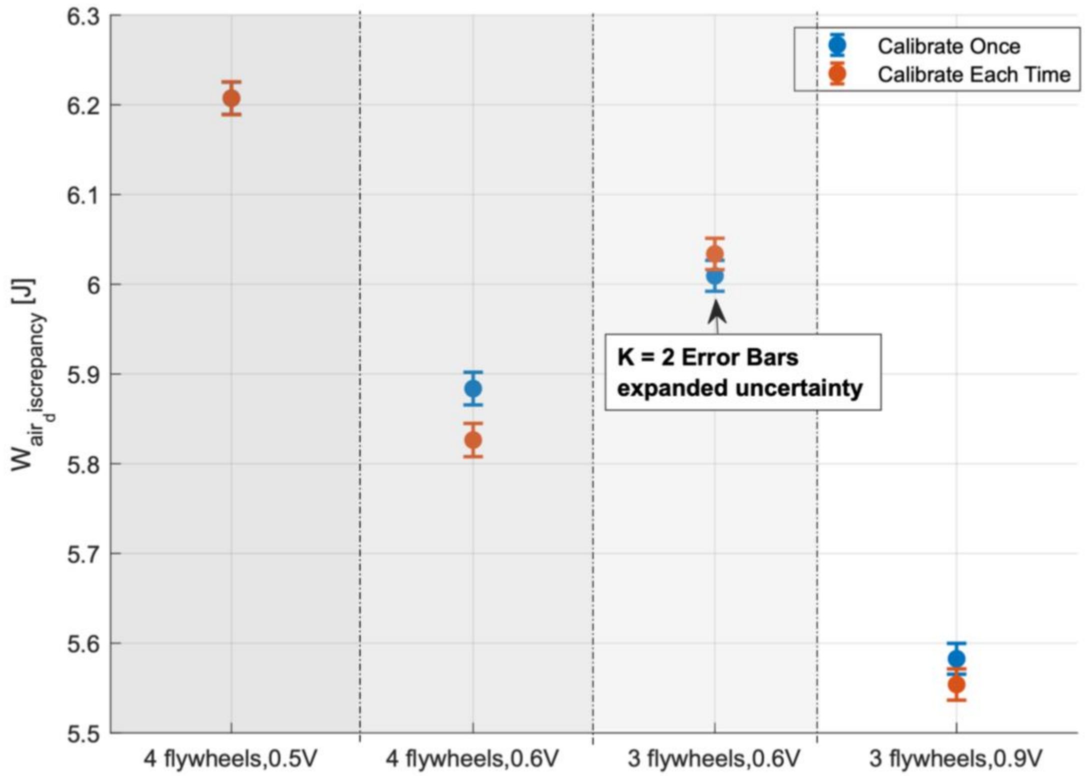

Figure 2: *Effect of calibration method on theoretical vs experimental discrepancy*

Therefore, an intermittent calibration procedure (one calibration per every three runs) is recommended to balance the tradeoff of calibration time with the known hysteresis effects of the force transducer.

Friction Mapping and Model Selection

Friction mapped against engine speed exhibits a steeper slope at low−mid speeds with a soft knee near ~40 rad/s. A single global fit was selected for accounting, shown in Equation 4, with $D \approx 4.0 \times 10^{-3}\ N \cdot m$ and $b \approx 0.43\ N \cdot m$.

$$T_f\ [N \cdot m] = 0.004\omega \left[\frac{rad}{s}\right] + 0.4286 \quad [4]$$

As shown in Figure 3, although a two-segment fit reduces residuals above 40 rad/s, the global line slightly over-predicts low-speed loss and under-predicts mid-band loss, yielding conservative net work estimates across the usable window. If future use targets a narrow speed band, a piecewise fit will decrease uncertainty.

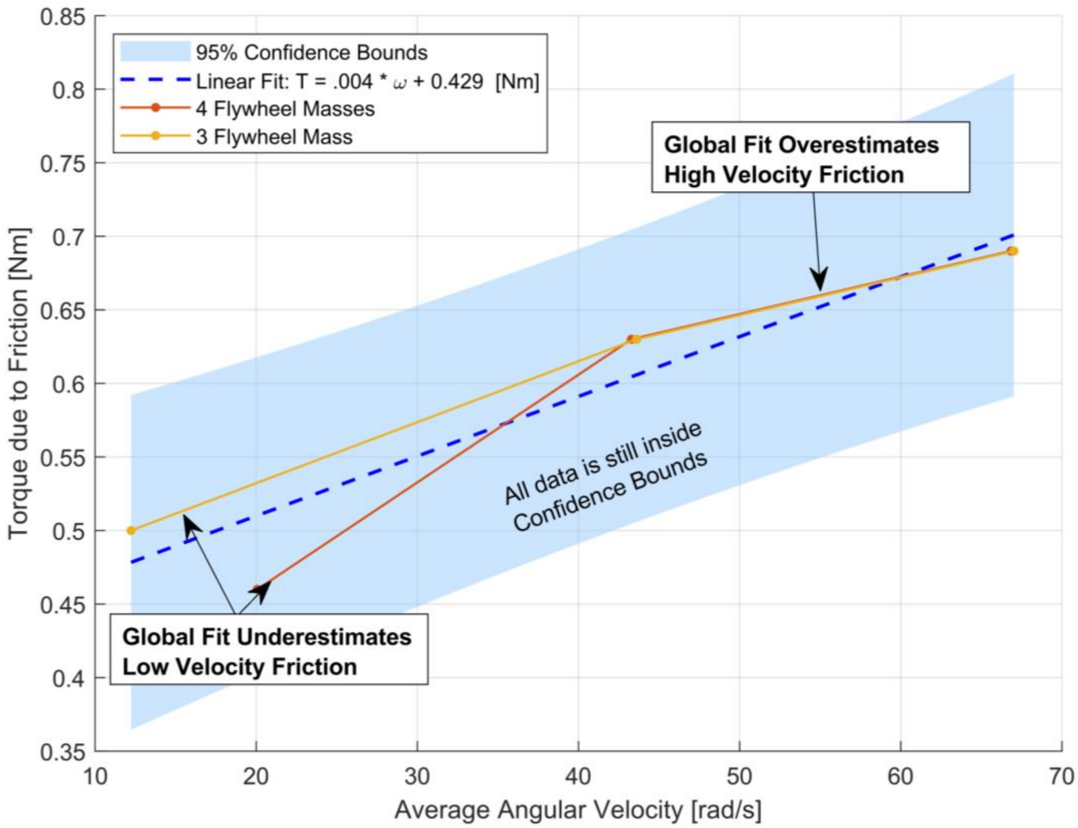

Figure 3: *Friction torque T_f vs. speed ω for 3 and 4 flywheel stacks with linear fit*

Friction Model Validation and Flywheel Inertia

To validate the friction model, the inertia of one flywheel was calculated using Equations 2a and 2b. Data was collected with four and three flywheels respectively and subtracted to find the inertia of a single flywheel. Higher flywheel inertia was chosen to be able to characterize the system at lower speeds without stalling. Using this method, the experimental inertia for one flywheel is $I_{\text{fly,exp}} = 2.80 \times 10^{-3} \pm 0.0021\ \text{kg} \cdot \text{m}^2$, compared with the geometry estimate $I_{\text{fly,theortical}} = 2.60 \times 10^{-3} \pm 0.0021\ \text{kg} \cdot \text{m}^2$. The resulting 7–8% positive bias is consistent with (i) the conservative friction model slightly overestimating dissipated work at low–mid speeds, (ii) additional rotating hardware (hubs, fasteners, keys) not included in the plate-only calculation, and (iii) small calibration/quantization effects in torque and speed. Experimental values for all four flywheel settings were found using the friction model and used in future calculations.

Theoretical vs Experimental Discrepancies

A significant discrepancy exists between the theoretical and experimental calculations of work done by air (no injection). This discrepancy is most clearly highlighted in trials without air injection, allowing for isolation of the W_{air} term and comparison to the theoretical value. The energy balance in Equation 2 assumes perfect air sealing, adiabatic compression and expansion, and negligible air flow resistance. As shown in Figure 4, the discrepancy between the experimental and theoretical value decreases as the control voltage increases (speed increases), suggesting the main source of the discrepancy is due to unmodeled leakage effects.

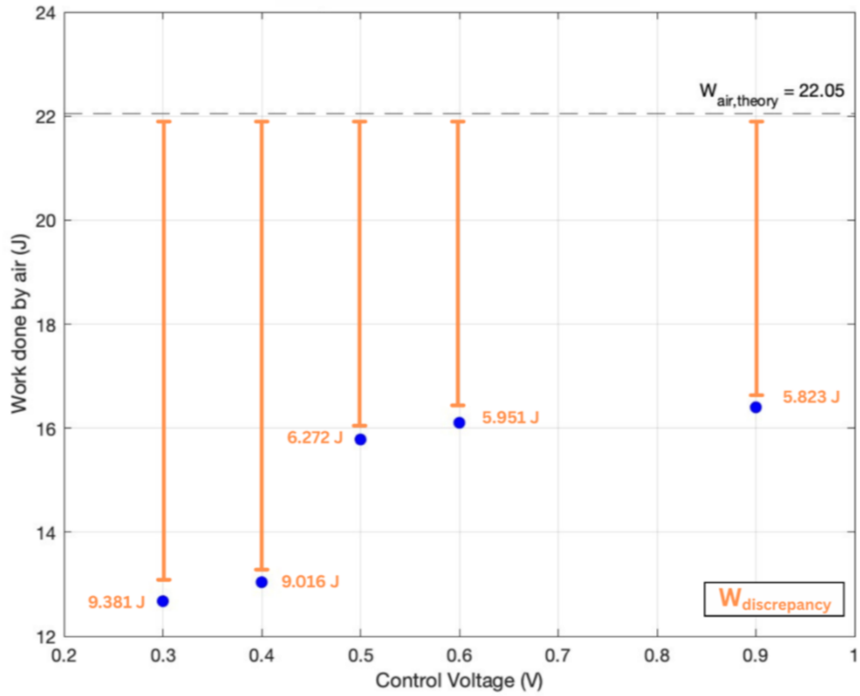

Figure 4: *Discrepancy between theoretical and experimental work of air for different control voltages with 4 flywheels*

At lower speeds, the duration of the compression stroke increases, allowing more time for air to leak out of the system. However, the discrepancy begins to plateau at higher control voltages (higher speeds). Therefore, control voltages greater than 0.5 V with four flywheels are recommended to maximize power and minimize leakage effects.

After testing for a prolonged period, the engine temperature was observed to decrease and the exhaust started to freeze, as shown in Figure 5. Not only does this invalidate the adiabatic assumption, but it also provides physical evidence that the seal effectiveness decreases due to decreased temperatures. To maximize power and align with the theoretical model, an improved sealing method is recommended.

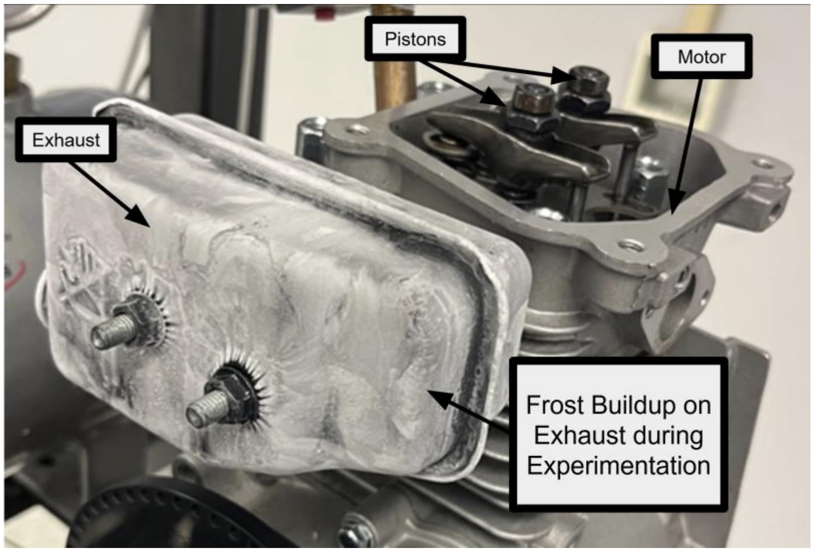

Figure 5: *Frost buildup on the exhaust due to unmodeled heat transfer effects*

However, even with the unmodeled leakage effects, a representative cycle (no injection) broken down per stroke still follows the expected behavior, as shown in Figure 6. During compression, work is done to compress the volume of air within the cylinder (-16.11 ± 0.01 J). During expansion, the work done should be equal and opposite to the work done in the compression stroke. By nature, the power stroke cannot do more work than the compression stroke. Due to lack of air injection and leakage, the power stroke does significantly less work than the compression stroke (3.42 ± 0.01 J). Because the cylinder is isobaric with the atmosphere during intake and exhaust, no work should be done those strokes. However, there is a small discrepancy suggesting further unmodeled effects such as resistance to air flow through the valve and heat transfer.

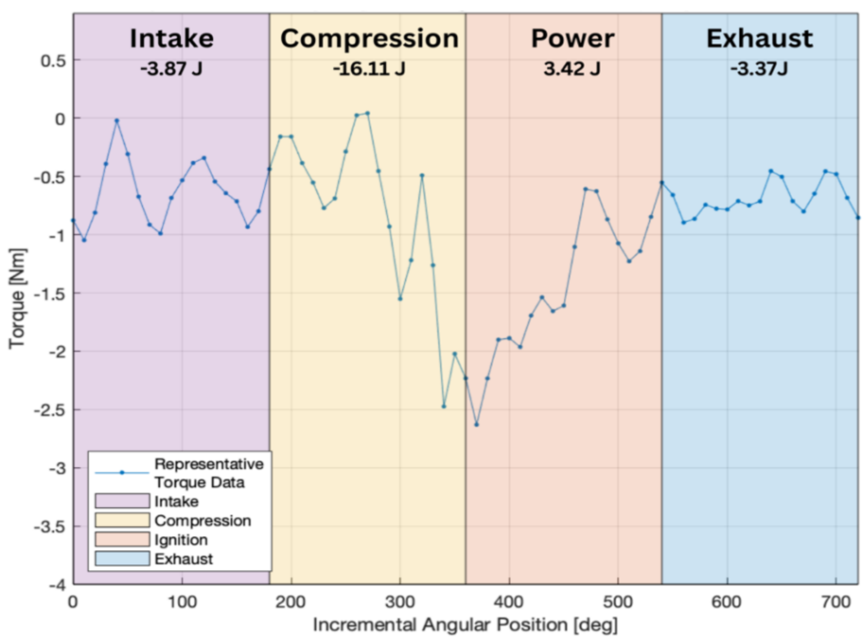

Figure 6. *Representative cycle with W_{air} over a 720° cycle for 0.6 V and 4 Flywheels with shaded strokes (Intake, Compression, Ignition, Exhaust).*

Maximum Power Output and Injection Timing Sweep

An optimal motor configuration using four flywheels, injection range of 355°-45°, and control voltage of 0.7 V achieved the highest power output of ~110.5 W. All testing was completed with four flywheels and voltages of 0.7V and 0.9V. These settings were chosen over lower voltage settings due to their higher inertia and speeds, lowering the possible effects of stalling. For 3–4 flywheels, reliable non-stall operation was observed only at ≥ 0.6 V (best configuration at 0.7 V). For 1-2 flywheels, only the 0.9 V condition sustained non-stall operation. The angle notation refers to 355°, 5° *before* the power stroke (TDC = 0°), and 45° into the power stroke. To further maximize power, the air injection start angle and duration were iterated at 0.7 V to determine an optimal injection window, as shown in Figure 7.

Start Angle (degrees from Top Dead Center)

End Angle (degrees from TDC)	330	345	0	15	30	45	60
15	-113.5	62.4	-114.2				
30	-36.9	94.3	83.8	-115.4			
45	77.4	94.1	**102.0**	28.2	-115.8		
60	92.2	94.5	82.0	33.7	-28.0	-117.8	
75	77.4	83.8	85.0	34.1	-30.8	-83.4	-121.5
90	76.4	75.4	75.7	19.7	-30.8	-76.5	-92.6

Best

Worst

Start Angle (degrees from TDC)

End Angle (degrees from TDC)	345	350	355	0
30	94.3	87.9	91.5	83.8
35	89.3	100.2	100.6	94.3
40	87.0	102.0	107.0	99.3
45	94.1	100.5	**110.5**	102.0
50	100.5	105.1	100.9	92.5
55	95.9	90.3	106.8	92.2
60	94.5	102.2	100.6	82.0
65	80.7	87.6	89.8	92.4

Best

Better

Good

Figure 7: *Injection timing map for 0.7 V: average mechanical power vs. start/stop angles; recommended window 0°→45° window (~102.0 W) (left); further refinement 355°→45° window (~110.5 W) (right)*

Injection parameters were also evaluated at other settings, such as 0.9 V; the resultant power calculations were substantially lower than that of the 0.7V settings, as shown in Figure 8.

Start Angle (degrees from TDC)

End Angle (degrees from TDC)	315	330	345	0	15
15	61.3	59.2	-46.5	N/A	
30	56.7	64.9	44.8	-23.6	N/A
45	N/A	-23.0	N/A	N/A	-3.1

Good

Bad

Figure 8: *Injection timing map for 0.9 V: average mechanical power vs. start/stop angles; recommended 330°→30° window (~64.9 W), but further testing may be performed to fine-tune these bounds if needed.*

A start angle greater than 0° is too late in the power stroke to provide a meaningful increase in power generation; in fact, due to the air needing to travel a distance between when the controller registers TDC and when the air injection hits the motor, start angles of before 0° performed better, since the injection hits closer to TDC. However, as shown by comparing Figures 7 and 8, increasing the control voltage, and thus the engine speed, results in diminishing returns. Even though there are more cycles at higher control voltages, each cycle is less efficient, supporting the choice of 0.7 V to maximize power.

Performance Limitations and Further Improvements

Discrepancies between theoretical and experimental work values exposed unmodeled system effects, primarily air leakage. Testing confirmed that leakage is most pronounced during the compression stroke at slower speeds. Given these findings, Burdell Inc. recommends enhancing the piston and valve sealing mechanisms to minimize air leakage. To further improve the discrepancies between theoretical and experimental work values, the friction model should be updated with a fit per flywheel configuration. This will decrease the uncertainty propagated in the model, resulting in more accurate work calculations.

Further sensor and equipment upgrades can make identification and control more efficient. On the identification side, cleaner low-frequency speed estimation (e.g., Savitzky–Golay filtering), 5° binning with per-cycle micro-shift, and explicit TDC/cam sensing reduce alignment and quantization error; logging actual injector on-time tightens the mapping from command to delivered work. On the control side, a power setpoint scheduler that incrementally adapts dwell maintains a commanded net power across operating conditions, and a two-stage injection (small pre-window near TDC followed by a main window) improves repeatability close to TDC without raising peak load. If future revisions allow, a higher-resolution encoder and in-cylinder pressure tracing would further reduce uncertainty and directly validate the air-spring assumption underpinning the timing map.

Burdell Inc recommends improving the seals, accuracy of the friction model, and sensor quality to further Webb Industries' goal of efficient generator operation and alignment with the theoretical model.

Conclusion

Angle-injection identification produced a generator-ready configuration for Webb Industries' compressed-air IC engine. The torque scale $T = G\,V_{ft} + T_0$ ($G \approx 0.717,\ T_0 \approx -0.003\ N \cdot m$) and Newtonian friction model $T_f(\omega) = 0.4286 + 0.0040\,\omega$ (rad/s) support stroke-level accounting within K=2 bands. Enabling pneumatic injection identifies a robust cross-TDC window (≈355°→45°) that repeatedly delivers ~110.5 W mechanical at 0.7 V. Recommended operation is $\omega \approx 90–110$ rad/s (860 – 1050 rpm) with the 4-flywheel stack for reliable power output. With sealing improvement, the configuration is ready for generator integration and provides a clear path to repeatable ~110.5 W output with margin to measured limits.